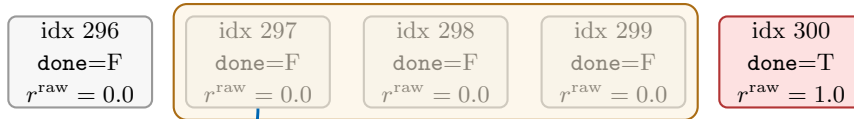


1. Raw stream: one 1-step transition per `rb.add()` call, in temporal order



window used to resolve idx **297** (arrives complete once idx 299 streams in)

2. What actually gets WRITTEN to buffer storage (verified against `offline.buffer_cache/b7bfa8d3`)

idx	stored next.reward	nonterminal	stored gamma	steps_to next_obs
297	0.000000	True	0.99252	3
298	0.995006	False	0.99252	3
299	0.997500	False	0.99501	2
300	1.000000	False	0.99750	1

FIFO delay =
 $n-1 = 2$ calls

Per-transition formula — `MultiStepTransform` (`rb.transforms.py`), configured with $n=3$ steps, $\gamma = 0.9975$ (this run's `algo.n_step / algo.gamma`):

window for transition starting at t is $[t, t+1, t+2]$. Let k = number of steps until a `next.done=True` frame is reached (capped at $n-1 = 2$ if no `done` appears in the window).

$$\begin{aligned} \text{stored next.reward} &= \sum_{i=0}^k \gamma^i r_{t+i}^{\text{raw}} & \text{stored gamma} &= \gamma^{k+1} \\ \text{steps_to_next_obs} &= k+1 & \text{nonterminal} &= (k=n-1) \wedge \neg(\text{done at } t+k) \end{aligned}$$

idx 297: no `done` anywhere in $[297, 298, 299] \Rightarrow k=2$ (full reach), so **nonterminal=True** — this transition bootstraps $\gamma^3 \cdot Q(\text{obs}_{300})$ at training time (its own accumulated reward is 0, all three raw rewards in its window are 0).

idx 298: window $[298, 299, 300]$ still reaches the FULL $k=2$ steps ahead (raw rewards at 298,299 are both 0, only 300's raw reward is 1) $\Rightarrow$ reward = $\gamma^2 \cdot 1 = 0.995006$. But **nonterminal=False** anyway — not because the window was cut short, but because the frame it lands on (300) is *itself* `done=True`: $(k=n-1)$ is true, yet $\neg(\text{done at } t+k)$ is false. (*idx 299 is the other case: window would reach 301, but done hits already at 300 — one step short — so $k=1 < n-1$, capped short instead.*)

idx 300: *is itself* a `done=True` frame $\Rightarrow k=0$: reward = $r_{300}^{\text{raw}} = 1.0$ exactly, **nonterminal=False** $\Rightarrow$ TD target = 1.0 exactly, no bootstrap term at all.